

DSA Patterns Roadmap (Topic Wise)

Interview preparation ke liye major DSA topics aur unke common patterns.

Arrays

- Two Pointers
- Sliding Window
- Prefix Sum
- Kadane's Algorithm
- Merge Intervals
- Cyclic Sort

Strings

- Sliding Window
- Hashing
- Two Pointers
- KMP
- Rabin-Karp
- Z Algorithm

Linked List

- Fast & Slow Pointer
- Reverse Linked List
- Cycle Detection
- Dummy Node

Stack

- Monotonic Stack
- Next Greater Element
- Histogram Pattern
- Parentheses Matching

Queue & Deque

- BFS
- Monotonic Queue
- Sliding Window Maximum

Binary Search

- Classic Binary Search
- Lower/Upper Bound
- Binary Search on Answer

Recursion & Backtracking

- Pick/Not Pick
- Subsets
- Permutations
- N-Queens

Trees

- DFS
- BFS
- Diameter
- LCA
- Path Sum
- Tree DP

BST

- Inorder Property
- Validation
- Kth Smallest
- Successor/Predecessor

Heap

- Top-K
- K-way Merge
- Running Median
- Greedy + Heap

Graphs

- DFS
- BFS
- Topological Sort
- Union Find
- Dijkstra
- MST

Greedy

- Interval Scheduling
- Activity Selection
- Sort + Pick Best

Dynamic Programming

- Knapsack
- LCS
- LIS
- Grid DP
- Interval DP
- Stock DP
- Bitmask DP

Trie

- Prefix Search
- Word Dictionary
- XOR Problems

Bit Manipulation

- XOR
- Bitmasking
- Subset Generation
- Power of Two

Segment Tree / Fenwick Tree

- Range Query
- Range Update
- Lazy Propagation

Maths

- GCD/LCM
- Sieve
- Modular Arithmetic
- Fast Exponentiation

DSU

- Connected Components

- Cycle Detection
- Dynamic Connectivity

Recommended Learning Order

1. Arrays
2. Strings
3. Linked List
4. Stack
5. Queue
6. Binary Search
7. Recursion
8. Backtracking
9. Trees
10. BST
11. Heap
12. Graphs
13. Greedy
14. Dynamic Programming
15. Trie
16. Bit Manipulation
17. Segment Tree
18. Maths
19. DSU